

Hypertherm®

XPR™

Preventive maintenance program



809490 – REVISION 2

ENGLISH



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XPR

Instruction Manual

809490
REVISION 2

English
Original instructions

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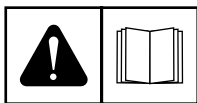
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For training and education resources, go to the Hypertherm Cutting Institute (HCI) online at www.hypertherm.com/hci.



ENGLISH

WARNING! Before operating any Hypertherm equipment, read the safety instructions in your product's manual, the *Safety and Compliance Manual* (80669C), *Waterjet Safety and Compliance Manual* (80943C), and *Radio Frequency Warning Manual* (80945C). Failure to follow safety instructions can result in personal injury or in damage to equipment.

Copies of the manuals can come with the product in electronic and printed formats. Electronic copies are also on our website. Many manuals are available in multiple languages at www.hypertherm.com/docs.

BG (БЪЛГАРСКИ/BULGARIAN)

ПРЕДУПРЕЖДЕНИЕ! Преди да работите с което и да е оборудване Hypertherm, прочетете инструкциите за безопасност в ръководството на вашия продукт, „Инструкция за безопасност и съответствие“ (80669C), „Инструкция за безопасност и съответствие на Waterjet“ (80943C) и „Инструкция за предупреждение за радиочестота“ (80945C).

Копия на ръководствата може да придружават продукта в електронен и в печатен формат. Можете да получите копия на ръководствата, предлагани на всички езици, от „Documents library“ (Библиотека за документи) на адрес www.hypertherm.com.

CS (ČESKY/CZECH)

VAROVÁNÍ! Před uvedením jakéhokoli zařízení Hypertherm do provozu si přečtěte bezpečnostní pokyny v příručce k produktu a v *Manuálu pro bezpečnost a dodržování předpisů* (80669C), *Manuálu pro bezpečnost a dodržování předpisů při řezání vodním paprskem* (80943C) a *Manuálu varování ohledně rádiových frekvencí* (80945C).

Kopie příruček a manuálů mohou být součástí dodávky produktu, a to v elektronické i tištěné formě. Kopie příruček a manuálů ve všech jazykových verzích, v nichž byly dané příručky a manuály vytvořeny, naleznete v „Knihovně dokumentů“ na webových stránkách www.hypertherm.com.

DA (DANSK/DANISH)

ADVARSEL! Inden Hypertherm udstyr tages i brug skal sikkerhedsinstruktionerne i produktets manual og i *Manual om sikkerhed og overholdelse af krav* (80669C), *Manual om sikkerhed og overholdelse af krav for vandstråleskæring* (80943C), og *Manual om radiofrekvensadvarsel* (80945C), gennemlæses.

Kopier af manualerne kan ledsage produktet i elektroniske og trykte formater. Du kan også få kopier af manualer, på alle sprog der er til rådighed for hver manuel, fra „Dokumentbiblioteket“ på www.hypertherm.com.

DE (DEUTSCH/GERMAN)

WARNUNG! Bevor Sie ein Hypertherm-Gerät in Betrieb nehmen, lesen Sie bitte die Sicherheitsanweisungen in Ihrer Bedienungsanleitung, das *Handbuch für Sicherheit und Übereinstimmung* (80669C), das *Handbuch für Sicherheit und Compliance bei Wasserstrahl-Schneidanlagen* (80943C) und das *Handbuch für Hochfrequenz-Warnung* (80945C).

Bedienungsanleitungen und Handbücher können dem Gerät in elektronischer Form oder als Druckversion beiliegen. Alle Handbücher und Anleitungen können in den jeweils verfügbaren Sprachen auch in der „Dokumente-Bibliothek“ unter www.hypertherm.com heruntergeladen werden.

ES (ESPAÑOL/SPANISH)

¡ADVERTENCIA! Antes de operar cualquier equipo Hypertherm, lea las instrucciones de seguridad del manual de su producto, del *Manual de seguridad y cumplimiento* (80669C), del *Manual de seguridad y cumplimiento en corte con chorro de agua* (80943C) y del *Manual de advertencias de radiofrecuencia* (80945C).

Pueden venir copias de los manuales en formato electrónico e impreso junto con el producto. También se pueden obtener copias de los manuales, en todos los idiomas disponibles para cada manual, de la “Biblioteca de documentos” en www.hypertherm.com.

ET (EESTI/ESTONIAN)

HOIATUS! Enne Hyperthermi mis tahes seadme kasutamist lugege läbi toote kasutusjuhendis olevad ohutusjuhised ning *Ohutus- ja vastavusjuhend* (80669C), *Veejoa ohutuse ja vastavuse juhend* (80943C) ja *Raadiosageduse hoiatusjuhend* (80945C). Ohutusjuhiste eiramine võib põhjustada vigastusi ja kahjustada seadmeid.

Juhiste koopiad võivad olla tootega kaasas elektroonilises ja trükitvormingus. Juhiste koopiad kõigis iga käsiraamatu jaoks saadaolevas keeles saate hankida ka „Documents library (dokumentide raamatukogust)” lehel www.hypertherm.com.

FI (SUOMI/FINNISH)

VAROITUS! Ennen minkään Hypertherm-laitteen käyttöä lue tuotteen käyttöoppaassa olevat turvallisuusohjeet, *turvallisuuden ja vaatimustenmukaisuuden käsikirja* (80669C), *vesileikkauksen turvallisuuden ja vaatimustenmukaisuuden käsikirja* (80943C) ja *radiotaajuusvaroitusten käsikirja* (80945C).

Käyttöoppaiden kopiot voivat olla tuotteen mukana elektronisessa ja tulostetussa muodossa. Voit saada käyttöoppaiden kopiot kaikilla kielillä ”latauskirjastosta”, joka on osoitteessa www.hypertherm.com.

FR (FRANÇAIS/FRENCH)

AVERTISSEMENT! Avant d'utiliser tout équipement Hypertherm, lire les consignes de sécurité du manuel de votre produit, du *Manuel de sécurité et de conformité* (80669C), du *Manuel de sécurité et de conformité du jet d'eau* (80943C) et du *Manuel d'avertissement relatif aux radiofréquences* (80945C).

Des copies de ces manuels peuvent accompagner le produit en format électronique et papier. Vous pouvez également obtenir des copies de chaque manuel dans toutes les langues disponibles à partir de la « Bibliothèque de documents » sur www.hypertherm.com.

GR (ΕΛΛΗΝΙΚΑ/GREEK)

ΠΡΟΕΙΔΟΠΟΙΗΣΗ! Πριν θέσετε σε λειτουργία οποιονδήποτε εξοπλισμό της Hypertherm, διαβάστε τις οδηγίες ασφαλείας στο εγχειρίδιο του προϊόντος και στο *εγχειρίδιο ασφαλείας και συμμόρφωσης* (80669C), στο *εγχειρίδιο ασφαλείας και συμμόρφωσης του waterjet* (80943C) και στο *εγχειρίδιο προειδοποιήσεων για τις ραδιοσυχνότητες* (80945C).

Αντίγραφο των εγχειριδίων μπορεί να συνοδεύουν το προϊόν σε ηλεκτρονική και έντυπη μορφή. Μπορείτε, επίσης, να λάβετε αντίγραφο των εγχειριδίων σε όλες τις γλώσσες που διατίθενται για κάθε εγχειρίδιο από την ψηφιακή βιβλιοθήκη εγγράφων (Documents library) στη διαδικτυακή τοποθεσία www.hypertherm.com.

HU (MAGYAR/HUNGARIAN)

VIGYÁZAT! Mielőtt bármilyen Hypertherm berendezést üzemeltetne, olvassa el a biztonsági információkat a termék kézikönyvében, a *Biztonsági és szabálykövetési kézikönyvben* (80669C), a *Vízugaras biztonsági és szabálykövetési kézikönyvben* (80943C) és a *Rádiófrekvenciás figyelmeztetéseket tartalmazó kézikönyvben* (80945C).

A termékhez a kézikönyv példányai elektronikus és nyomtatott formában is mellékelve lehetnek. A kézikönyvek példányai (minden nyelven) a www.hypertherm.com weboldalon a „Documents library” (Dokumentum könyvtár) részben is beszerezhetők.

ID (BAHASA INDONESIA/INDONESIAN)

PERINGATAN! Sebelum mengoperasikan peralatan Hypertherm, bacalah petunjuk keselamatan dalam manual produk Anda, *Manual Keselamatan dan Kepatuhan* (80669C), *Manual Keselamatan dan Kepatuhan Jet Air* (80943C), dan *Manual Peringatan Frekuensi Radio* (80945C). Kegagalan mengikuti petunjuk keselamatan dapat menyebabkan cedera pribadi atau kerusakan pada peralatan.

Produk mungkin disertai salinan manual dalam format elektronik maupun cetak. Anda juga dapat memperoleh salinan manual, dalam semua bahasa yang tersedia untuk setiap manual, dari "Perpustakaan dokumen" di www.hypertherm.com.

IT (ITALIANO/ITALIAN)

AVVERTENZA! Prima di usare un'attrezzatura Hypertherm, leggere le istruzioni sulla sicurezza nel manuale del prodotto, nel *Manuale sulla sicurezza e la conformità* (80669C), nel *Manuale sulla sicurezza e la conformità Waterjet* (80943C) e nel *Manuale di avvertenze sulla radiofrequenza* (80945C).

Il prodotto può essere accompagnato da copie elettroniche e cartacee del manuale. È anche possibile ottenere copie del manuale, in tutte le lingue disponibili per ogni manuale, dall'“Archivio documenti” all'indirizzo www.hypertherm.com.

JA (日本語/JAPANESE)

警告！Hypertherm 機器を操作する前に、この製品説明書にある安全情報、「安全とコンプライアンスマニュアル」(80669C)、「ウォータージェット の安全とコンプライアンス」(80943C)、「高周波警告」(80945C)をお読み ください。

説明書のコピーは、電子フォーマット、または印刷物として製品に同梱さ れています。各説明書は、www.hypertherm.com の「ドキュメントライブラ リ」から各言語で入手できます。

KO (한국어/KOREAN)

경고! Hypertherm 장비를 사용하기 전에 제품 설명서와 안전 및 규정 준수 설명서(80669C), 워터젯 안전 및 규정 준수 설명서(80943C) 그리고 무선 주파수 경고 설명서(80945C)에 나와 있는 안전 지침을 읽으십시오.

전자 형식과 인쇄된 형식으로 설명서 사본이 제품과 함께 제공될 수 있습니다. www.hypertherm.com 의 'Documents library (문서 라이브러리)' 에서도 모든 언어로 이용할 수 있는 설명서 사본을 얻을 수 있습니다.

NE (NEDERLANDS/DUTCH)

WAARSCHUWING! Lees voordat u Hypertherm-apparaat gebruikt de veiligheidsinstructies in de producthandleiding, in de *Veiligheids- en nalevingshandleiding* (80669C) in de *Veiligheids- en nalevingshandleiding voor waterstralen* (80943C) en in de *Waarschuwingshandleiding radiofrequentie* (80945C).

De handleidingen kunnen in elektronische en gedrukte vorm met het product worden meegeleverd. De handleidingen, elke handleiding beschikbaar in alle talen, zijn ook verkrijgbaar via de “Documentenbibliotheek” op www.hypertherm.com.

NO (NORSK/NORWEGIAN)

ADVARSEL! Før du bruker noe Hypertherm-utstyr, må du lese sikkerhetsinstruksjonene i produktets håndbok, *håndboken om sikkerhet og samsvar* (80669C), *håndboken om vannjet sikkerhet og samsvar* (80943C), og *håndboken om radiofrekvensadvarsler* (80945C).

Eksemplarer av håndbøkene kan medfølge produktet i elektroniske og trykte utgaver. Du kan også få eksemplarer av håndbøkene i alle tilgjengelige språk for hver håndbok fra dokumentbiblioteket på www.hypertherm.com.

PL (POLSKI/POLISH)

OSTRZEŻENIE! Przed rozpoczęciem obsługi jakiegokolwiek systemu firmy Hypertherm należy się zapoznać z instrukcjami bezpieczeństwa zamieszczonymi w podręczniku produktu, w *podręczniku bezpieczeństwa i zgodności* (80669C), *podręczniku bezpieczeństwa i zgodności systemów strumienia wody* (80943C) oraz *podręczniku z ostrzeżeniem o częstotliwości radiowej* (80945C).

Do produktu mogą być dołączone kopie podręczników w formacie elektronicznym i drukowanym. Kopie podręczników, w każdym udostępnionym języku, można również znaleźć w „Bibliotece dokumentów” pod adresem www.hypertherm.com.

PT (PORTUGUÊS/PORTUGUESE)

ADVERTÊNCIA! Antes de operar qualquer equipamento Hypertherm, leia as instruções de segurança no manual do seu produto, no *Manual de Segurança e de Conformidade* (80669C), no *Manual de Segurança e de Conformidade do Waterjet* (80943C) e no *Manual de Advertência de radiofrequência* (80945C).

Cópias dos manuais podem acompanhar os produtos nos formatos eletrônico e impresso. Também é possível obter cópias dos manuais em todos os idiomas disponíveis para cada manual na “Biblioteca de documentos” em www.hypertherm.com.

RO (ROMÂNĂ/ROMANIAN)

AVERTIZARE! Înainte de utilizarea oricărui echipament Hypertherm, citiți instrucțiunile de siguranță din manualul produsului, *manualul de siguranță și conformitate* (80669C), *manualul de siguranță și conformitate Waterjet* (80943C) și din *manualul de avertizare privind radiofrecvența* (80945C).

Produsul poate fi însoțit de copii ale manualului în format tipărit și electronic. De asemenea, dumneavoastră puteți obține copii ale manualelor, în toate limbile disponibile pentru fiecare manual, din cadrul secțiunii „Biblioteca documente” aflată pe site-ul www.hypertherm.com.

RU (РУССКИЙ/RUSSIAN)

БЕРЕГИСЬ! Перед работой с любым оборудованием Hypertherm ознакомьтесь с инструкциями по безопасности, представленными в руководстве, которое поставляется вместе с продуктом, в *Руководстве по безопасности и соответствию* (80669C), в *Руководстве по безопасности и соответствию для водоструйной резки* (80943C) и *Руководстве по предупреждению о радиочастотном излучении* (80945C).

Копии руководств, которые поставляются вместе с продуктом, могут быть представлены в электронном и бумажном виде. Копии руководств на всех языках, на которые переведено то или иное руководство, можно также загрузить в разделе «Библиотека документов» на веб-сайте www.hypertherm.com.

SK (SLOVENČINA/SLOVAK)

VÝSTRAHA! Pred použitím akéhokoľvek zariadenia od spoločnosti Hypertherm si prečítajte bezpečnostné pokyny v návode na obsluhu vášho zariadenia a v *Manuáli o bezpečnosti a súlade s normami* (80669C), *Manuáli o bezpečnosti a súlade s normami pre systém rezania vodou* (80943C) a v *Manuáli s informáciami o rádiových frekvenciách* (80945C).

Kópia návodu, ktorá je dodávaná s produktom, môže mať elektronickú alebo tlačенú podobu. Kópie návodov, vo všetkých dostupných jazykoch, sú k dispozícii aj v sekcii z „knžnice Dokumenty“ na www.hypertherm.com.

SL (SLOVENŠČINA/SLOVENIAN)

OPOZORILO! Pred uporabo katerekoli Hyperthermove opreme preberite varnostna navodila v priročniku vašega izdelka, v *Priročniku za varnost in skladnost* (80669C), v *Priročniku za varnost in skladnost sistemov rezanja z vodnim curkom* (80943C) in v *Priročniku Opozorilo o radijskih frekvencah* (80945C).

Izdelku so lahko priloženi izvodi priročnikov v elektronski ali tiskani obliki. Izvode priročnikov v vseh razpoložljivih jezikih si lahko prenesete tudi iz knjižnice dokumentov “Documents library” na naslovu www.hypertherm.com.

SR (SRPSKI/SERBIAN)

UPOZORENJE! Pre rukovanja bilo kojom Hyperthermovom opremom pročitajte uputstva o bezbednosti u svom priručniku za proizvod, *Priručniku o bezbednosti i usaglašenosti* (80669C), *Priručniku o bezbednosti i usaglašenosti Waterjet tehnologije* (80943C) i *Priručniku sa upozorenjem o radio-frekvenciji* (80945C).

Može se dogoditi da kopije priručnika prate proizvod u elektronskom i štampanom formatu. Takođe možete da pronadete kopije priručnika, na svim jezicima koji su dostupni za svaki od priručnika, u “Biblioteci dokumenata” (“Documents library”) na www.hypertherm.com.

SV (SVENSKA/SWEDISH)

VARNING! Läs häftet säkerhetsinformationen i din produkts *säkerhets- och efterlevnadsmanual* (80669C), *säkerhets- och efterlevnadsmanualen för Waterjet* (80943C) och *varningsmanualen för radiofrekvenser* (80945C) för viktig säkerhetsinformation innan du använder eller underhåller Hypertherm-utrustning.

Kopior av manualen kan medfölja produkten i elektronisk och tryckform. Du hittar även kopior av manualerna i alla tillgängliga språk i dokumentbiblioteket (Documents library) på www.hypertherm.com.

TH (ภาษาไทย/THAI)

คำเตือน! ก่อนการใช้งานอุปกรณ์ของ Hypertherm ทั้งหมด โปรดอ่านคำแนะนำด้านความปลอดภัยในคู่มือการใช้สินค้า คู่มือด้านความปลอดภัยและการปฏิบัติตาม (80669C), คู่มือด้านความปลอดภัยและการปฏิบัติตามสำหรับการใช้หัวตัดระบบวอเตอร์เจ็ต (80943C) และ คู่มือคำเตือนเกี่ยวกับความถี่วิทยุ (80945C) การไม่ปฏิบัติตามคำแนะนำด้านความปลอดภัยอาจส่งผลให้เกิดการบาดเจ็บหรือเกิดความเสียหายต่ออุปกรณ์

สินค้าอาจมีสำเนาคู่มือในรูปแบบอิเล็กทรอนิกส์และแบบสิ่งพิมพ์แนมมาด้วย นอกจากนี้ คุณสามารถขอรับสำเนาคู่มือแต่ละประเภทเป็นภาษาต่าง ๆ ที่มีให้ใช้งานได้ที่ “คลังเอกสาร” ในเว็บไซต์ www.hypertherm.com

TR (TÜRKÇE/TURKISH)

UYARI! Bir Hypertherm ekipmanını çalıştırmadan önce, ürününüzün kullanım kılavuzunda, *Güvenlik ve Uyumluluk Kılavuzu'nda* (80669C), *Su Jeti Güvenlik ve Uyumluluk Kılavuzu'nda* (80943C) ve *Radyo Frekansı Uyarısı Kılavuzu'nda* (80945C) yer alan güvenlik talimatlarını okuyun.

Kılavuzların kopyaları, elektronik ve basılı formatta ürünle birlikte verilebilir. Her biri tüm dillerde yayınlanan kılavuzların kopyalarını www.hypertherm.com adresindeki “Documents library” (Dosyalar kitaplığı) başlığından da elde edebilirsiniz.

VI (TIẾNG VIỆT/VIETNAMESE)

CẢNH BÁO! Trước khi vận hành bất kỳ thiết bị Hypertherm nào, hãy đọc các hướng dẫn an toàn trong hướng dẫn sử dụng sản phẩm của bạn, *Sổ tay An toàn và Tuân thủ* (80669C), *Sổ tay An toàn và Tuân thủ Tia nước* (80943C), và *Hướng dẫn Cảnh báo Tần số Vô tuyến* (80945C). Không tuân thủ các hướng dẫn an toàn có thể dẫn đến thương tích cá nhân hoặc hư hỏng thiết bị.

Bản sao của các hướng dẫn sử dụng có thể đi kèm sản phẩm ở định dạng điện tử và bản in. Bạn cũng có thể lấy bản sao của các hướng dẫn sử dụng, thuộc tất cả các ngôn ngữ hiện có cho từng hướng dẫn sử dụng, từ “Thư viện tài liệu” tại địa chỉ www.hypertherm.com.

ZH-CN (简体中文/CHINESE SIMPLIFIED)

警告！在操作任何海宝设备之前，请阅读产品手册、《安全和法规遵守手册》(80669C)、《水射流安全和法规遵守手册》(80943C) 以及《射频警告手册》(80945C) 中的安全操作说明。

随产品提供的手册可能提供电子版和印刷版两种格式。您也可从“Documents library”（文档资料库）中获取每本手册所有可用语言的副本，网址为 www.hypertherm.com。

ZH-TW (繁體中文/CHINESE TRADITIONAL)

警告！在操作任何 Hypertherm 設備前，請先閱讀您產品手冊內的安全指示，包括《安全和法規遵從手冊》(80669C)、《水刀安全和法規遵從手冊》(80943C)，以及《無線電頻率警示訊號手冊》(80945C)。

手冊複本可能以電子和印刷格式隨附產品提供。您也可以在此 www.hypertherm.com 的「文檔資料庫」內獲取所有手冊的多語種複本。

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Preventive Maintenance Program

Overview

Congratulations on the purchase of your Hypertherm plasma system.

Hypertherm plasma systems can operate in harsh conditions for many years. To maintain system performance, minimize operating costs, and lengthen system life, follow the maintenance procedures given in this Preventative Maintenance Program (PMP).

Hypertherm made this PMP specifically for your plasma system. The PMP has two parts: a cleaning and inspection schedule and a component replacement schedule.

If you have questions about how to maintain your plasma system, contact your OEM or regional Hypertherm Technical Service team. You can find contact information for each regional office at www.hypertherm.com on the “Contact us” page.

This document refers to your system’s instruction manual. If you do not have your instruction manual, you can find it at www.hypertherm.com/docs.

- *XPR170 Instruction Manual: 810060*
- *XPR300 Instruction Manual: 809480*



The graphics in this document are for reference only. The appearance and location of your cutting system parts and web interface screens can be different, based on the type of XPR cutting system that you have.

Cleaning and inspection schedule

This is a daily, weekly, and monthly schedule for cleaning and inspections. The PMP has instructions for each task. These instructions help your personnel understand what to do and what to look for during each task. In general, an operator can do the daily and weekly tasks, while maintenance personnel usually do the monthly tasks.

A monthly maintenance log is in the back of this manual. You can make photocopies of the maintenance log to record tasks.

Table 1

Maintenance task or activity	Daily	Weekly	Monthly	Every 6 months
Do a test of the inlet pressures	X			
Examine all of the air filters	X			
Do a check of the coolant level and condition	X			
Examine and lubricate O-rings	X			
Examine the water tube and torch	X			
Examine hoses, cables, and leads		X		
Do tests for gas leaks		X		
Do a check of the coolant flow		X		
Clean inside the plasma power supply			X	
Examine the contactors			X	
Examine the pilot arc relay			X	
Examine the coolant system			X	
Do the coolant flow test			X	
Examine the gas line connections			X	
Examine the hoses			X	
Examine the cables			X	
Examine the ground connections			X	
Examine the table-to-workpiece connection			X	
Replace the coolant and coolant filter, and clean and examine the pump screen				X

Daily

Do a test of the inlet pressures

Adjust the gas pressures dynamically, with gas flowing. Refer to [Table 2](#). It is best to use the pressure transducer output from the XPR cutting system. You can use the gas pressures shown in the XPR web interface. Refer to your system's instruction manual. To show the gas pressures on the CNC, contact your cutting machine supplier.

Test gas pressures in preflow, cutflow, and pierceflow mode with gas flowing. The correct pressure is based on the gas type for the process found in the cut charts. Refer to the *XPR Cut Charts Instruction Manual* (809830).

Table 2 – Gas quality, pressure, and flow requirements

Gas	Quality	System inlet pressure (during gas flow*)	Flow rate
O ₂ (oxygen)	99.5% pure, clean, dry, oil-free	Core, VWI: 7.5 bar $\pm$ 0.4 (110 psi $\pm$ 5) OptiMix: 7.9 bar $\pm$ 0.4 (115 psi $\pm$ 5)	71 slpm (150 scfh)
N ₂ (nitrogen)	99.99% pure, clean, dry, oil-free	Core, VWI: 7.5 bar $\pm$ 0.4 (110 psi $\pm$ 5) OptiMix: 8.3 bar $\pm$ 0.4 (120 psi $\pm$ 5)	181 slpm (380 scfh)
Air**	Clean, dry, oil free consistent with 8573-1:2010 Class 1.4.2	Core, VWI: 7.5 bar $\pm$ 0.4 (110 psi $\pm$ 5) OptiMix: 7.9 bar $\pm$ 0.4 (115 psi $\pm$ 5)	118 slpm (250 scfh)
H ₂ (hydrogen)	99.995% pure	OptiMix: 8.3 bar $\pm$ 0.4 (120 psi $\pm$ 5)	50 slpm (105 scfh)
Ar (argon)	99.99% pure; clean, dry, oil-free	VWI: 7.5 bar $\pm$ 0.4 (110 psi $\pm$ 5) OptiMix: 8.3 bar $\pm$ 0.4 (120 psi $\pm$ 5)	118 slpm (250 scfh)
F5 (95% nitrogen, 5% hydrogen)	99.98% pure	VWI: 7.5 bar $\pm$ 0.4 (110 psi $\pm$ 5) OptiMix: 7.9 bar $\pm$ 0.4 (115 psi $\pm$ 5)	40 slpm (85 scfh)

* When there is **no** gas flow, make sure that the pressure at the gas inlet connection is less than 8.6 bar (125 psi) to avoid system alerts.

** Any air compressors that supply air to the cutting system must remove oil prior to air delivery.

Hypertherm recommends that air compressors supply air that obey the following requirements of ISO Standard 8573-1:2010 Class 1.4.2:

Maximum particle count in 1.0 m³:

- 20,000 at 0.1 microns – 0.5 microns
- 400 at 0.5 microns – 1.0 microns
- 10 at 1.0 microns – 5.0 microns

Maximum water vapor pressure dew point:

3°C (37°F)

Maximum oil concentration:

0.1 mg/m³ (for aerosol, liquid, and vapor)

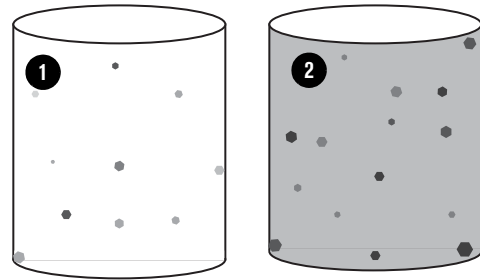


Speak to your air compressor manufacturer if you operate the cutting system in temperatures colder than 3°C (37°F) or if you are unsure that the air compressor can obey the ISO standard for air quality.

Examine all of the air filters

Examine all of the air filters for contamination.

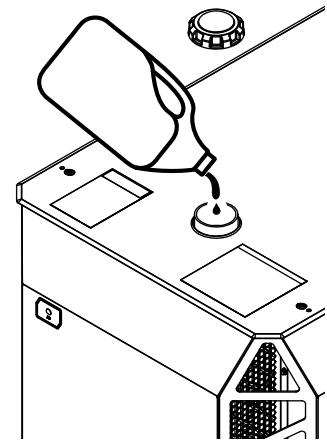
- If you find moisture replace the air filter.
- If you find a moderate quantity of solid contamination, such as dust, blow air through it or use suction to clean the filter. ❶
- If you find a large quantity of solid contamination, replace the air filter. ❷



Do a check of the coolant level and condition

Make sure that the reservoir is full:

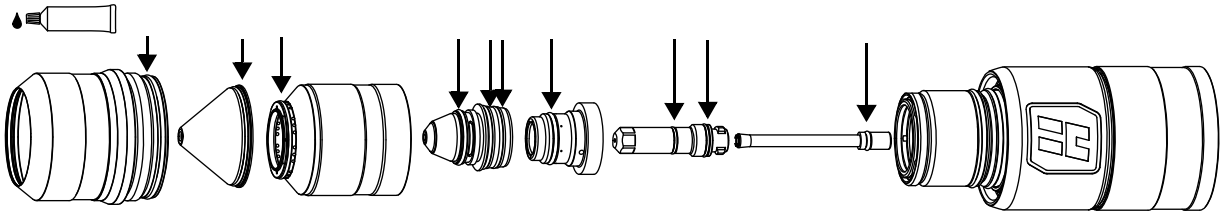
1. Remove the cap from the reservoir fill port inlet that is located on top of the plasma power supply.
2. look into the fill port inlet to see into the coolant reservoir.
3. If the level of the coolant is below the base of the fill spout, add coolant (028872) until the coolant level gets to the base of the fill spout.
4. Record the number of gallons your system requires in [Coolant requirements](#) on page 36.



Examine the coolant reservoir for contamination. If you find any contamination, then replace all of the coolant. Refer to [Replace the coolant and coolant filter, and clean and examine the pump screen](#) on page 24.

Examine and lubricate O-rings

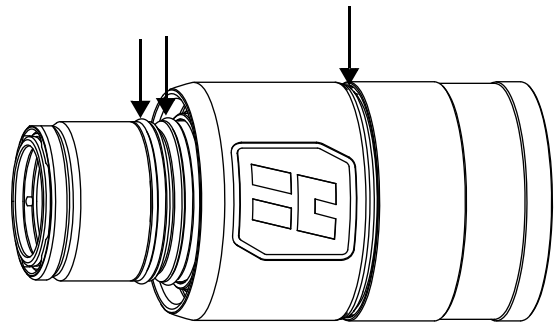
1. Remove the consumables from the torch.
2. Examine all of the O-rings on the torch and consumables for damage or excess wear.



3. If you find damage or excess wear, replace the O-ring.



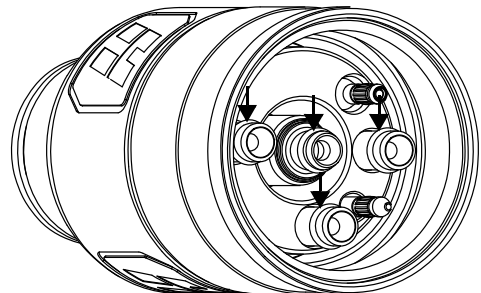
Receptacle O-rings and torch O-rings for the torch main body are available as part of the torch rebuild and filter kit (428639 or 428640). Other O-rings are included with the consumables.



4. Apply a thin film of silicone lubricant (027055) to all O-rings.



The O-rings should look shiny. Too much lubricant can prevent gas flow.



5. Install all O-rings so they fit snugly.
6. Examine all threaded consumables, and remove any dirt from the threads.
7. Continue to [Examine the water tube and torch](#) on page 16.

Examine the water tube and torch

1. Remove the used water tube ❶ from the torch.
2. Examine the water tube for bends and pitting. If you find bends or pitting, replace the water tube.
3. If you do not find bends or pitting on the water tube, examine the O-ring ❷ on the end of the water tube:
 - a. Replace the O-ring if you find damage or excess wear.



Torch rebuild kits are available from Hypertherm.

- b. Apply a thin film of silicone lubricant (027055) if replacement is not needed.

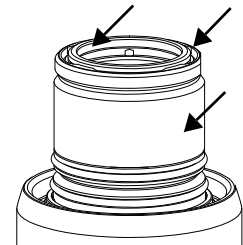
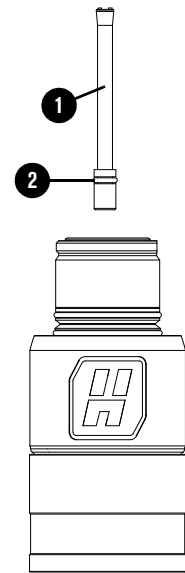


The O-ring should look shiny. Too much lubricant can restrict water tube motion. Remove excess lubricant if found.

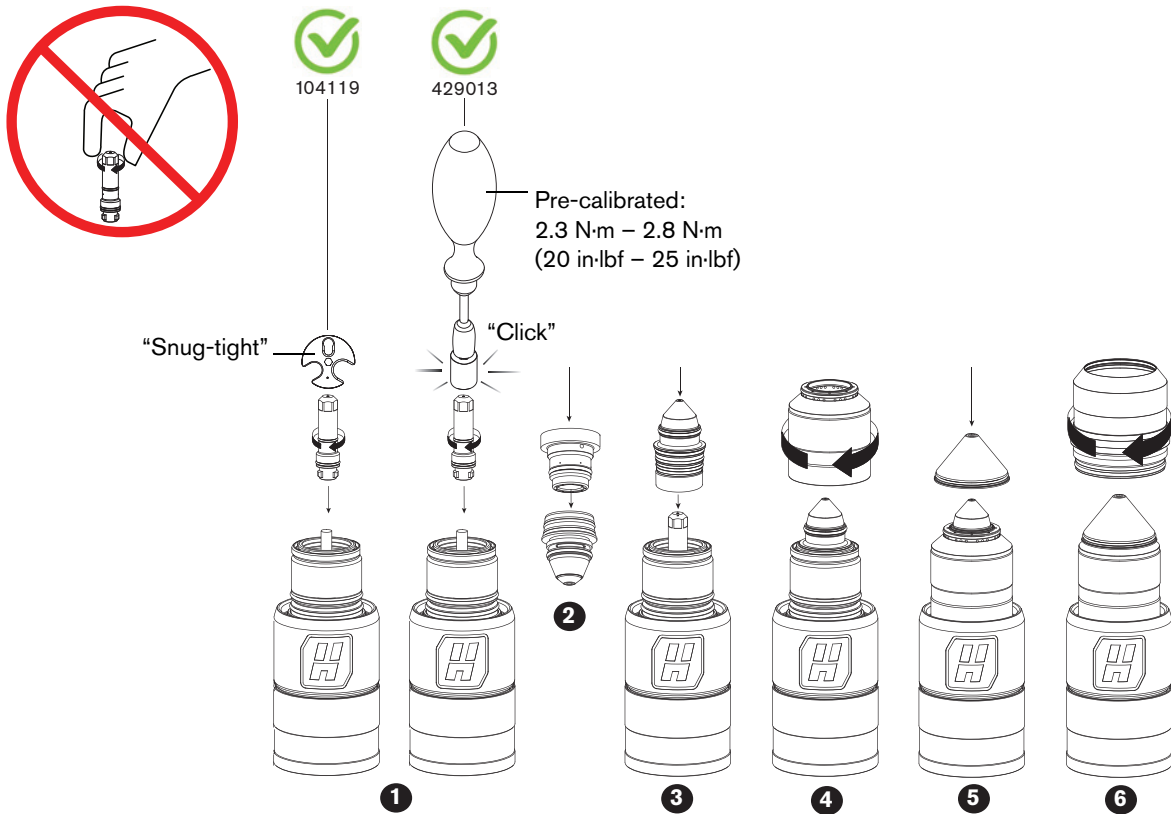
4. Examine the nozzle and electrode mating surfaces on the torch main body for damage or pitting. If found, replace the torch main body.
5. If you do not find damage or pitting, use a clean paper towel or lint-free cloth to clean the internal and external surfaces of the torch. Use a cotton swab to clean internal surfaces that are not easy to get to.



Make sure that you do not leave any cotton fibers on the inner surfaces of the torch main body.



6. Use compressed air to remove any remaining particles from the torch main body.
7. Install the consumables on the torch.



Weekly

Examine hoses, cables, and leads

1. Examine all hoses, cables, and leads that connect system components. Look for:
 - Scrapes, cuts, or holes
 - Chemical spills or burns
 - Kinks or bends
2. Replace any hose, cable, or lead if you find damage.



Refer to the *Parts List* section of your system's instruction manual.

3. Examine all of the fittings that connect the hoses, cables, and leads:
 - a. Tighten loose connections if found, but do not make the connections too tight.
 - b. Order a replacement hose, cable, or lead set if you find its fitting has damage or excess wear. Replacement hoses, cables, and leads are available from Hypertherm.



Individual fittings for external hoses, cables, and leads are **not** replaceable. If you find a problem with an external fitting, you must order a replacement hose, cable, or lead set (with integrated fitting).

4. Make sure that the hoses, cables, and leads do not twist or kink during torch movement and system operation. Adjust them if needed.

Do tests for gas leaks

1. Use the CNC screen or XPR web interface to select the command to do an automated gas leak test. Test results and information will appear in the error log.



For information about how to do this, see the instruction manual that came with your CNC.

2. See the error log for information and guidance about how to diagnose or troubleshoot a possible gas leak.

Do a check of the coolant flow

1. Look at the CNC or plasma power supply screen on the XPR web interface to identify the coolant flow rate.
2. Make sure that the coolant flow rate is acceptable:
 - Between 3.79 L/min – 11.36 L/min (1 gal/min – 3.0 gal/min) for an XPR170 cutting system.
 - Between 3.79 L/min – 9.46 L/min (1 gal/min – 2.5 gal/min) for an XPR300 cutting system.



If the flow rate is outside of the correct range, an internal obstruction can be the cause.



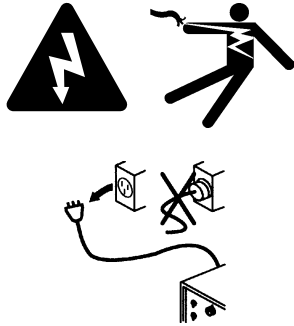
Shown for reference only.

Status

Type	XPR300-OptiMix
State	Wait for start
Log	0
Process	300A O2 Air
Arc Time	0a 0d 0h 16min 40s
(+) DC	300 A
Coolant Flow	7.96 lpm (2.1 gpm)
Coolant Level	Low

Monthly

WARNING



ELECTRIC SHOCK CAN KILL

Disconnect electric power before doing installation or maintenance. You can get a serious electric shock if electric power is not disconnected. Electric shock can seriously injure or kill you.

All work that requires removal of the plasma power supply outer cover or panels must be done by a qualified technician.

Refer to the *Safety and Compliance Manual* (80669C) for more safety information.

Clean inside the plasma power supply

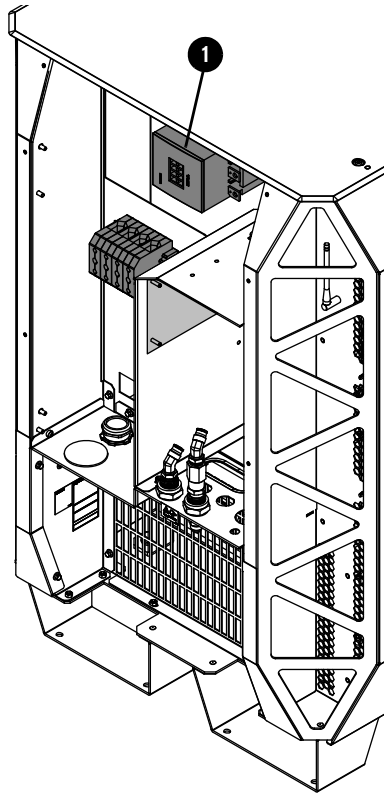
1. Remove the power from the cutting system:
 - a. Set the line-disconnect switch to the OFF position.
 - b. Make sure that the power-indicator LED is not illuminated on the plasma power supply, gas connect console, or torch connect console.
2. Remove the side panels of the plasma power supply and the top panel from both the gas connect console and TorchConnect™ console.
3. Use low pressure air or a vacuum to remove any accumulation of dust and particles from:
 - The side panels
 - The fans
 - The heat exchanger
 - The inside of the gas connect console
 - The inside of the TorchConnect console
4. Remove dust and particles from circuit boards.

 Be careful not to damage the circuit boards.
5. Install the side panels on the plasma power supply and top panel on both the gas connect console and TorchConnect console before you supply power to the cutting system.

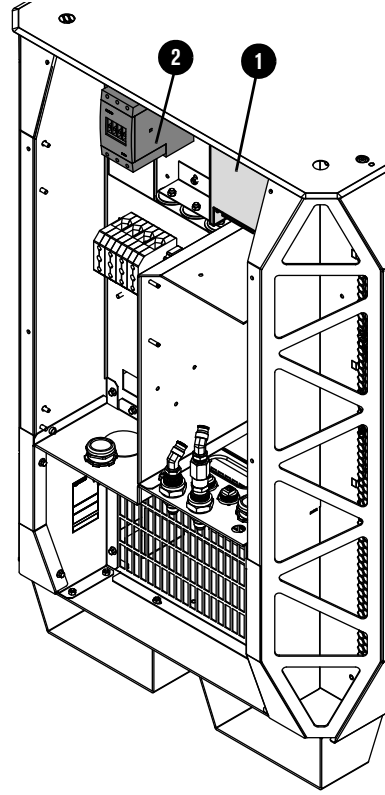
Examine the contactors

1. Examine the contacts for black or rough surfaces:

- For the XPR170 (left), examine the main contactor **1**.
- For the XPR300 (right), examine the main contactor **1** and inrush contactor **2**.



XPR170



XPR300

2. If you find too much of this condition, replace the contactor.

Examine the pilot arc relay

1. Remove the cover of the pilot arc relay.

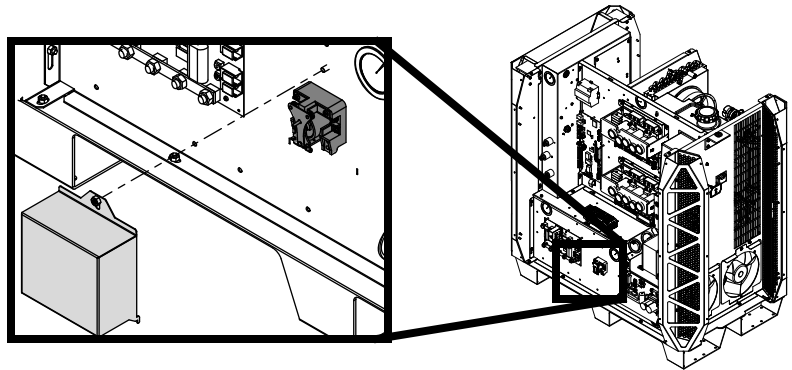


Shown for reference only.

2. Examine the contacts inside the relay for black or rough surfaces.
3. If you find too much of this condition, replace the pilot arc relay.
4. Install the cover of the pilot arc relay.



Be careful that you do not pinch the wires between the cover and the center panel.



Examine the coolant system

Examine the coolant system for coolant leaks at all connections. Make sure you examine:

- The gas connect console
- The TorchConnect console
- The torch main body
- The internal connection in the plasma power supply

Do the coolant flow test

1. Look at the CNC or plasma power supply screen on the XPR web interface to identify the coolant flow rate.



Shown for reference only.

2. Make sure that the coolant flow rate is acceptable:
 - Between 3.79 L/min – 11.36 L/min (1 gal/min – 3.0 gal/min) for an XPR170 cutting system.
 - Between 3.79 L/min – 9.46 L/min (1 gal/min – 2.5 gal/min) for an XPR300 cutting system.



If the flow rate is outside of the correct range, an internal obstruction can be the cause.

Status

Type	XPR300-OptiMix
State	Wait for start
Log	0
Process	300A O2 Air
Arc Time	0a 0d 0h 16min 40s
(+) DC	300 A
Coolant Flow	7.96 lpm (2.1 gpm)
Coolant Level	Low

3. If you find obstructions, remove them. Replace damaged parts if found.
4. If coolant flow is lower than 3.79 L/min (1 gal/min), and it has been more than 6 months since the last coolant replacement, replace the coolant. Refer to [Replace the coolant and coolant filter, and clean and examine the pump screen](#) on page 24.



Hypertherm recommends coolant replacement every 6 months.

5. If the coolant flow remains slow, refer to *How to measure coolant flow* in your system's instruction manual.
6. If you cannot find or resolve the problem with these corrective actions, contact your cutting machine supplier or regional Hypertherm Technical Service team.

Examine the gas line connections

Spray all of the gas line connections with soapy water. If bubbles appear on a gas line, tighten or replace it as necessary. Refer to the *Parts List* section of your system's instruction manual.

CAUTION

Do not clean brass connections with ammonia-based cleaners. Ammonia causes brass to crack and become brittle.

Examine the hoses

Examine each hose for kinks or sharp bends that can restrict gas flow or cause damage to the hose.

If the cutting table uses a power track system to support the leads that go from the power supply to the gas console or torch, look at the position of the leads in the track. Make sure the leads do not twist or kink. This can cause a restriction.

Examine the cables

Examine all cables for scratches or unusual wear. If the outside insulation is cut or has any other damage, replace the cable. Refer to the *Parts List* section of your system's instruction manual.

Examine the ground connections

Make sure that all components of the system are individually grounded to a driven earth ground. Refer to the *Installation and Grounding* section of your system's instruction manual.

Examine the table-to-workpiece connection

Examine the work lead (+) connection where the work lead (+) connects to the cutting table.

Make sure that there is no paint, oil, dirt, or rust on the workpiece. This type of contamination prevents a clean metal-to-metal contact between the work lead and the cutting table or workpiece. It can also cause arc-transfer problems.

Every 6 months

Replace the coolant and coolant filter, and clean and examine the pump screen

- 1.** Estimate the total coolant volume for your cutting system. Refer to *Estimate the total coolant volume for your cutting system* in your system's instruction manual.
- 2.** Remove the power from the cutting system.
- 3.** Remove the right external panel from the plasma power supply (this is the panel on the right when you look at the front of the unit).



M6 (10 mm hex) screws hold the panel in position.

- 4.** Remove old coolant from the coolant reservoir:
 - a.** Connect a 3/8-inch inner diameter tube to the outlet of the valve on the bottom of the reservoir.
 - b.** Put the other end of the tube into an empty container.

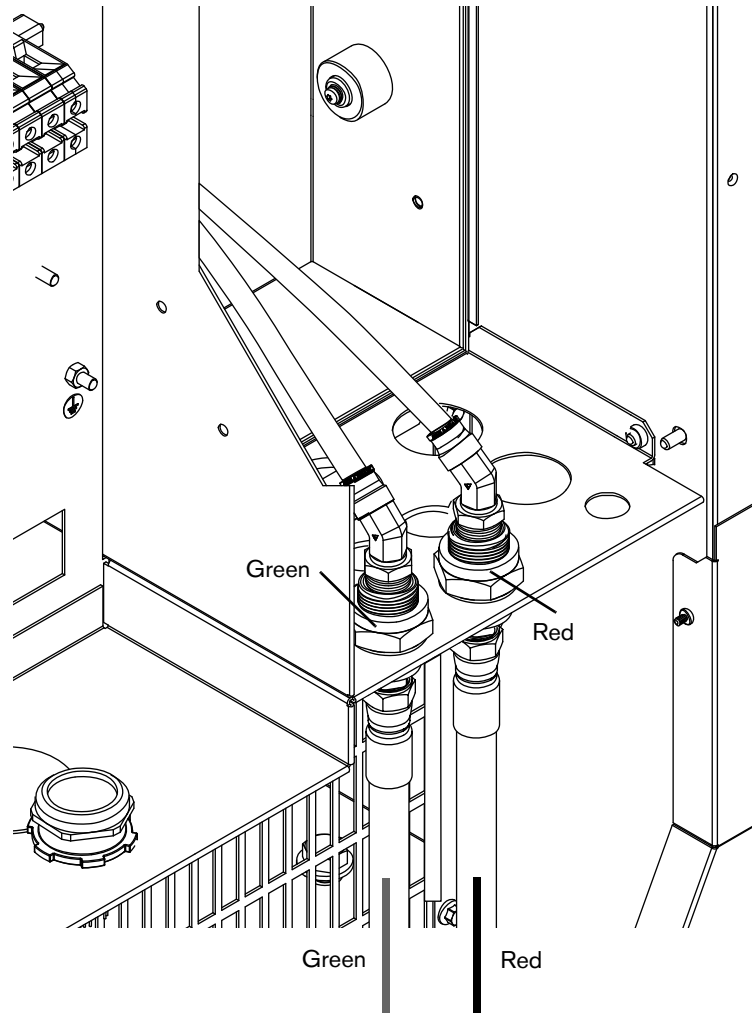


Use a container that holds the approximate total coolant volume for your cutting system.

- c.** Open the valve located on the bottom of the reservoir.
- d.** Remove the cap on the reservoir inlet to allow the coolant to flow out of the reservoir.

5. Remove old coolant from the heat exchanger:

- a.** Keep the 3/8-inch inner diameter tube connected to the outlet of the valve on the bottom of the reservoir on one end and the other end in the container.



- b.** Remove the coolant return hose (red band) from the rear of the plasma power supply.
- c.** Attach compressed air (no more than 6.89 bar/100 psi) to the coolant return hose fitting on the rear of the plasma power supply where the return coolant hose (red band) was previously connected.
- d.** For no more than 30 seconds, use the compressed air to blow all of the coolant back into to the reservoir and filter housing.



System components need the coolant to lubricate rotating surfaces. If air flows through the cutting system for longer than 30 seconds, it can eliminate the coolant necessary for lubrication.

- e.** Close the valve at the bottom of the reservoir and remove the 3/8-inch inner diameter tubing from the outlet.



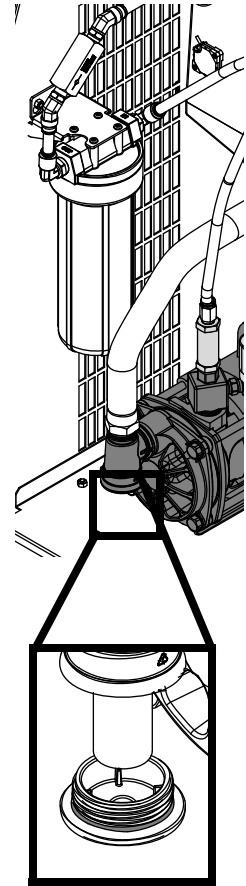
Do not store the 3/8-inch diameter tubing inside of the plasma power supply.

- f.** Leave the coolant return hose (red band) disconnected from the rear of the plasma power supply.
- g.** Put a container under the pump plug.
- h.** Remove the plug and coolant pump screen and set them aside.
- i.** Remove the coolant supply hose (green band) from the rear of the plasma power supply.
- j.** Attach compressed air (no more than 3.45 bar/50 psi) to the coolant supply hose fitting on the rear of the plasma power supply where the coolant supply hose (green band) was previously connected.
- k.** For no more than 30 seconds, use the compressed air to blow all of the coolant into the container.
- l.** Leave the coolant supply hose (green band) disconnected.



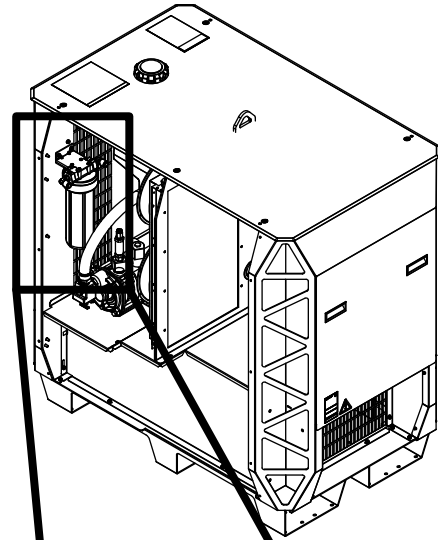
System components need the coolant to lubricate rotating surfaces. If air flows through the cutting system for longer than 30 seconds, it can eliminate the coolant necessary for lubrication.

- 6.** Clean and, if needed, replace the coolant pump screen:
 - a.** Clean the coolant pump screen. Rinse it with water if you find debris.
 - b.** Examine the coolant pump screen.
 - c.** If you find damage on the coolant pump screen, replace it (127559).
 - d.** Install the coolant pump screen.
 - e.** Wipe the O-ring on the plug. Make sure that the O-ring is free of debris, cracks, and nicks.
 - f.** Install the plug on the coolant pump housing.



7. Remove old coolant from the filter housing and replace the coolant filter:

- a.** Remove the filter housing from inside of the plasma power supply.
- b.** Discard all of the coolant from inside of the filter housing.
- c.** Remove and discard the coolant filter.
- d.** Examine the filter housing for debris. Rinse the filter housing to remove any debris, if found.
- e.** Install a new coolant filter (027005).
- f.** Install the filter housing.



8. Remove old coolant from hoses and leads:



Cutting system hoses and leads can hold a large volume of coolant.

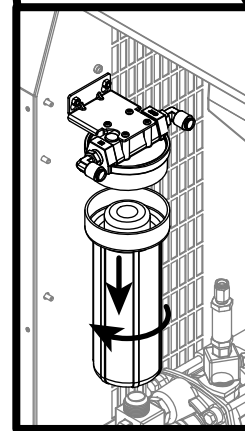


Make sure to remove all of the old coolant from the hoses and leads. If you do not, the new coolant will mix with the old coolant. This will cause the new coolant to degrade faster.

- a.** Put the disconnected end of the coolant return hose into an empty container.



Use a container that holds the approximate total coolant volume for your cutting system.



- b.** Attach compressed air (no more than 6.89 bar/100 psi) to the disconnected end of the coolant supply hose (green band).
- c.** For approximately 3 minutes, inject compressed air into the coolant supply hose fitting to force coolant out of the coolant return hose (red band) into an empty container.
- d.** After 3 minutes, look for coolant flow out of the coolant return hose (red band). Repeat this process until coolant flow from the coolant return hose (red band) stops.
- e.** When coolant flow from coolant return hose (red band) stops, connect both hoses to the rear of the plasma power supply.

9. Install new coolant. Refer to *How to fill the cutting system* with coolant in your system's manual.

Component replacement schedule

This is a list of components and their recommended replacement schedules. The schedule uses total arc hours to estimate when to replace a component. The use of cumulative arc hours is the most accurate method to estimate when you need to replace a component.

If you do not have a Hypertherm CNC and your CNC cannot track arc hours, use the guide below to estimate arc hours per year. This guide uses the average number of 8-hour shifts that the system operates on an average work day. For more information about calculating arc hours, contact your table manufacturer.

Average shifts per day	Estimated arc hours per year
1	500
2	1,000
3	1,500

Your cutting area and operations can have an effect on this schedule. Refer to [Cutting area and operation effects](#) on page 31.

Record information about your system and its replacement requirements in the following places:

- [System information](#) on page 36
- [Notes](#) on page 37

Table 3

Item Number	Quantity	Component	Cumulative number of arc hours											
			500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000
428639*	1	Kit: Torch rebuild and filter without coolant	X	X	X	X	X	X	X	X	X	X	X	X
428640*	1	Kit: Torch rebuild and filter with coolant	X	X	X	X	X	X	X	X	X	X	X	X
028872	6 – 12	Coolant (in US gallons)**	X	X	X	X	X	X	X	X	X	X	X	X
428878	1	Kit: Electronics***												
428879		200 V – 240 V (XPR170 only)		X		X		X		X		X		X
428641		380 V – 600 V (XPR170 only)												
428642		200 V – 240 V (XPR300 only)		X		X		X		X		X		X
428642		380 V – 600 V (XPR300 only)												
420220	1	Quick-disconnect torch receptacle				X				X				X
—†	1	Torch lead				X				X				X
006113	1	Coolant check valve						X						X
229821	2	Fan assembly, 120 mm (4.7 inch) diameter						X						X
229822	4	Fan assembly, 254 mm (10 inch) diameter						X						X
229741	1	Kit: Coolant flow meter						X						X
—†	1	Gas lead						X						X
—†	1	Pilot arc lead						X						X
141425	1	Power distribution PCB (PCB7)											X	
141354	1	Ignition PCB (PCB2)											X	
229838	1	High-frequency transformer (T1)												X
229640	1	Power source: 24 VDC											X	
229671	1	Power source: 48 VDC											X	
141368	1	Ohmic contact PCB (PCB2)											X	
141371	1	I/O PCB												X
229679	1	Chopper												X

- * This kit includes the air filter (011110), coolant filter (027005), and torch rebuild components.
- ** The number of gallons you need depends on the length of the leads. Refer to your system's instruction manual for more information, then record this number in [Coolant requirements](#) on page 36.
- *** Kit 428878 includes a pilot arc relay (003277), main contactor (003276), and torch main body (420221).
Kit 428879 includes a pilot arc relay (003277), main contactor (003294), and torch main body (420221).
Kit 428641 includes a pilot arc relay (003277), main contactor (003276), inrush contactor (229697), and torch main body (420221).
Kit 428642 includes a pilot arc relay (003277), main contactor (003294), inrush contactor (229697), and torch main body (420221).
Refer to your system's instruction manual for the correct voltage, then record this number in [System voltage](#) on page 36.
- † Part number varies. Refer to the *Parts List* in your system's instruction manual.

Cutting area and operation effects

The component replacement schedule is for reference only. Your cutting area conditions and operations can have an effect on your component replacement schedule.

These topics give the most common conditions and operations that increase the wear on some components in your plasma system. This wear decreases system performance and component life. If any of these conditions or operations apply, change your replacement schedule as suggested.

How much you need to change your schedule depends on how bad the condition is. If you have questions about your replacement schedule, contact your OEM or regional Hypertherm Technical Service team.

External cutting area

Consumables and torch assemblies stored in dirty cutting areas with no protection can collect contamination, such as dirt and metal dust. If this contamination gets into the torch or coolant, it can cause the following problems:

- Prevent O-rings and seals on the torch heads and receptacles from sealing
- Increase wear on torch heads and receptacles
- Increase wear on the coolant pump
- Cause unsatisfactory operation of the coolant flow switch



It can help to flush the coolant system or clean the flow switch. But, it can be necessary to replace the coolant switch to fully repair the operation of the coolant loop system.

The best solution is to store consumables and torch assemblies in a clean, protective space. Another solution is to use compressed air to clean these parts before you use them.

If you cannot store your consumables and torch heads in a protective space, you can replace the following components more frequently.

Description	Part number
Quick-disconnect torch receptacle	420220
Quick-disconnect torch	420221
Kit: Coolant pump and motor assembly	428729
Kit: Coolant flow meter	229741

Incoming air supply quality

To maintain system performance, it is important that the incoming air supply is clean. If the air supply has dirt, or water in it, components can become clogged or defective. An air supply of bad quality can cause the following problems:

- Decrease air flow
- Increase system gas errors
- Decrease cut quality and performance

Also, unwanted material in pressure sensors can cause false pressure readings.

The best solution, if possible, is to improve the quality of the air supply. Contact your OEM or regional Hypertherm Technical Service team if you need advice on how to improve the quality of your air supply.

If you cannot change your air supply quality, you can replace the following components more frequently.

Description	Part number
Air filter element	011110
Quick-disconnect torch	420221
Quick-disconnect torch receptacle	420220
Torch lead	—*
Gas solenoid valve, pressure transducer	—

* Part number varies

Oil in the torch head can cause a fire when the oil reacts with the oxygen cutting process. If you find oil in the air supply, contact your OEM or regional Technical Service team.

WARNING



FIRE HAZARD

Oil in the torch head can cause a fire.

Do not let oil into the air supply.

Metal dust inside the plasma power supply cabinet

Plasma cutting makes metal dust. If metal dust collects inside the plasma power supply, it can decrease the life of the fans.

The main power supply, chopper, or heat exchanger fans can operate more slowly, or failure of the fans can occur. This can cause temperature-related errors.

The best solution to extend the life of the fans is to clean inside the plasma power supply. Refer to [Clean inside the plasma power supply](#) on page 19.

If metal dust has collected inside your plasma power supply, you can replace the following components more frequently.

Description	Part number
Fan assembly: 292 cfm, 48 VDC, 120 mm (4.7 inch) diameter	229821
Fan assembly: 890 cfm, 48 VDC, 254 mm (10 inch) diameter	229822

Consumable use

If you let consumables reach complete failure, they can melt. Copper pieces can then break off and go into the coolant. In the coolant, these pieces can cause the following problems:

- Decrease coolant flow
- Increase coolant flow errors
- Cause damage to the consumables
- Cause coolant to leak at the check valve when you change the consumables
- Cause inaccurate coolant flow readings
- Decrease the life of the coolant pump
- Decrease the life of the check valve

When you do maintenance, examine the coolant filter for copper pieces. If you find copper pieces in the coolant filter, replace both the filter and coolant. If a coolant flow error occurs after you replace the filter and coolant, use the troubleshooting procedure in your system's instruction manual to find the correct action.

The best solution is to follow the usage guidelines for your consumables. However, if overuse occurs, you can replace the following components more frequently.

Description	Part number
Coolant	028872
Quick-disconnect torch receptacle	420220
Quick-disconnect torch	420221
Coolant filter (fine)	027005
Kit: Coolant pump and motor assembly	428729
Coolant check valve	006113
Kit: Coolant flow meter	229741

Cut-cycle time

When the cut-cycle is very short, relays operate more frequently. The system also pierces more frequently. Examples of this type of cycle are when you cut many small holes or make markings for numbers and letters. These types of operations can cause the following problems:

- Increase wear on pilot arc relay contact pads
- Increase wear on the starting components, such as the high-frequency transformer and the high-frequency ignition PCB
- Increase misfires and error codes

If your cut-cycle is short, you can replace the following components more frequently.

Description	Part number
Pilot arc relay	003277
Main contactor (200 V – 240 V)	003276
Main contactor (380 V – 600 V)	429060
Quick-disconnect torch	420221
High-frequency, high-voltage transformer (T1)*	229838
High-frequency, high-voltage ignition PCB (PCB2)*	141354

* These parts are in the gas connect console.

System information

Model number

Serial number

System voltage

- ☐ 200 V/208 V
- ☐ 220 V
- ☐ 240 V
- ☐ 380 V (CCC)
- ☐ 400 V (CE)
- ☐ 415 V (CE)
- ☐ 440 V
- ☐ 480 V (CSA)
- ☐ 600 V (CSA)

Coolant requirements

- ☐ 22.7 L (6 gal)
- ☐ 26.5 L (7 gal)
- ☐ 30.3 L (8 gal)
- ☐ 34.1 L (9 gal)
- ☐ 37.9 L (10 gal)
- ☐ 41.6 L (11 gal)
- ☐ 45.4 L (12 gal)
- ☐ Other:

Notes

[illegible]

